Brain Computer Interface

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BCI Applications

- Controlling robots

BCI Applications

- Controlling drones

BCI Applications

- BCI with VR

BCI Applications

- BCI speller

BCI Applications

- BCI mouse

BCI Applications

- BCI exoskeleton

BCI Applications

- NeuroPhone

BCI Applications

- Brain signal-based authentication system
- Epilepsy seizure detection
- Meditation applications
- Educational applications
- Emotion classification
- Sleep disorder studies
- etc.

Biosignals

- Signals that can be continually measured and monitored from the living being
- Can be electrical or non-electrical
  - Electrocardiogram (ECG)
  - Electroencephalogram (EEG)
  - Magnetoencephalogram (MEG)
  - Electrocorticography (ECoG)
  - Electromyogram (EMG)
  - Electrooculography (EOG)
  - Galvanic skin response (GSR)
  - Mechanomyogram (MMG)

What is BCI

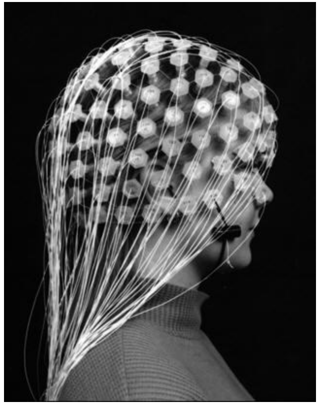
- Direct communication between enhanced brain and an external device.
- Methods
  - Invasive
  - Partially invasive
  - Non-invasive

Related subject areas

- Neuroscience
- Digital Signal Processing (DSP)
- Digital Image Processing (DIP)
- Machine Learning (ML) - in the most cases

Technologies

- fMRI: functional Magnetic Resonance Imaging
- EEG: Electroencephalography
- MEG: Magnetoencephalography
- NIRS: Near-infrared spectroscopy

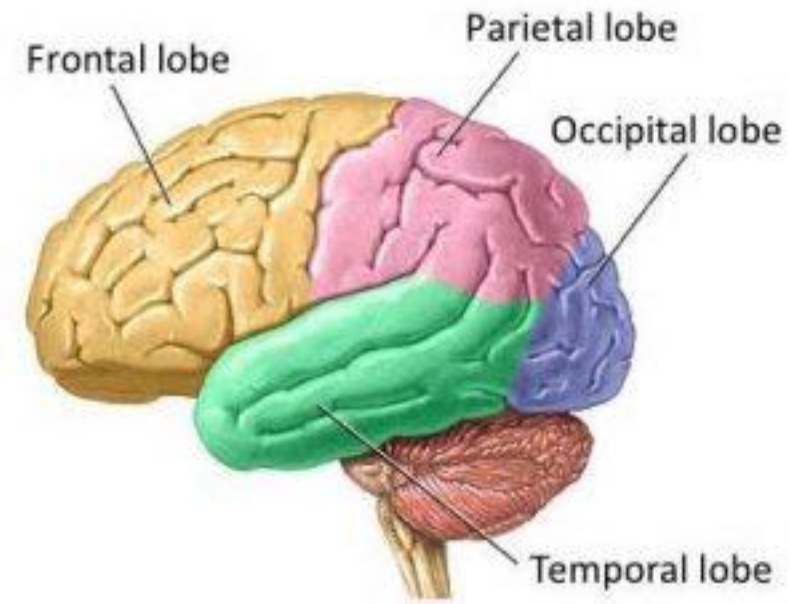



Basics of EEG

- Electroencephalography (EEG): Electrophysiological monitoring method to record electrical activity of one’s brain
- Subtle electric field is generated by neurons
- EEG electrodes can capture the electric field of group of neurons

Brain lobes

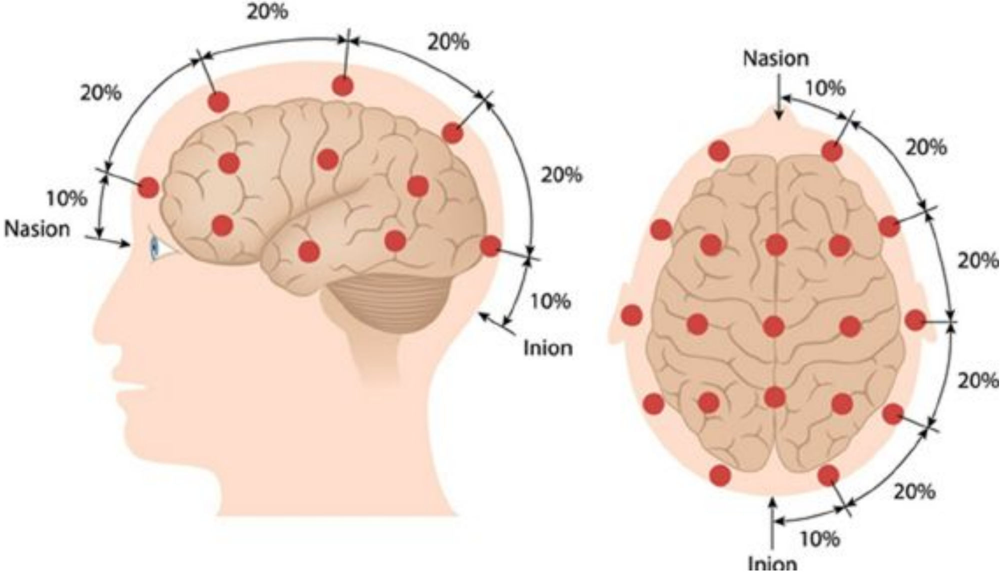
- Four brain lobes
  - Frontal
  - Parietal
  - Temporal
  - Occipital



Electrode placement

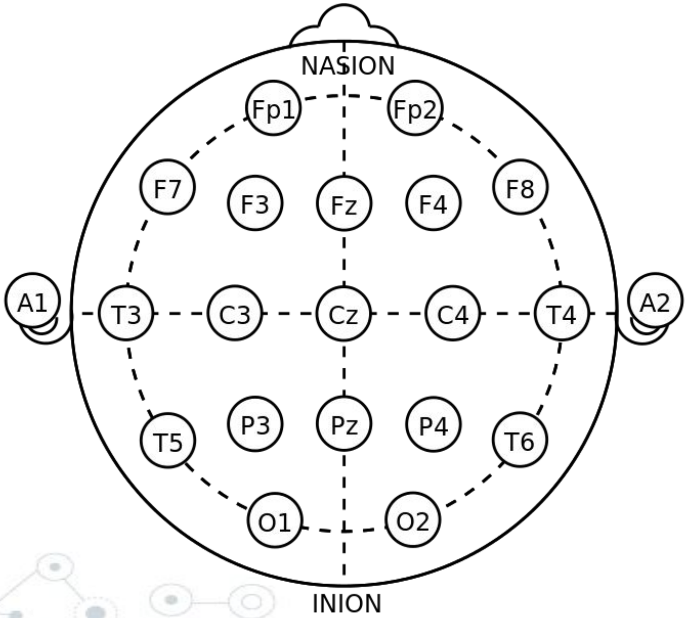
- 10-20 system
- 10-10 system or 10%
- 10-5 system

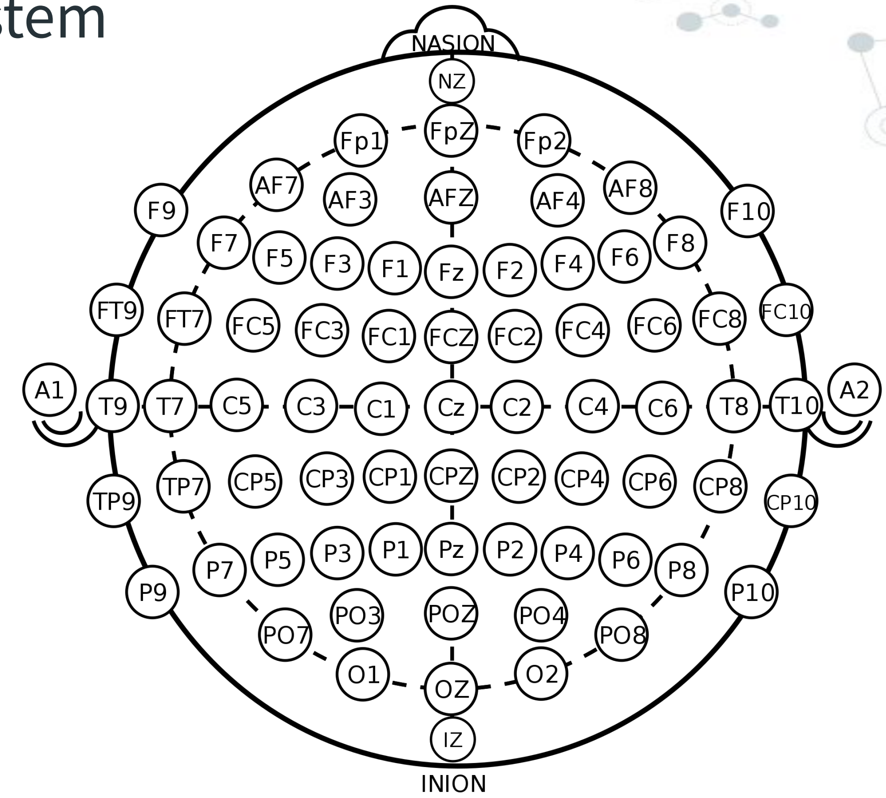
EEG Electrode Placement



Electrode placement (Cont.)

- 10-20 system & 10% system





Basic flow

Data Collection → **Pre-Processing** → **Feature Extraction** → **Classification**

EEG devices

- Clinical- and consumer-grade devices are available
- Two types of electrodes, Wet and dry
- Comes with different number of electrodes

EEG devices (Cont.)

Device	Number of Channels	Electrode type
NeuroSky Mindwave	1	Dry
Muse	2	Dry
EMOTIV Insight	5	Dry
EMOTIV Epoc+	14	Wet
OpenBCI	16	Dry
mBrainTrain	24	Wet
ENOBIO 32	32	Dry
Cognionics Mobile- 72	64	Dry

Pre-Processing

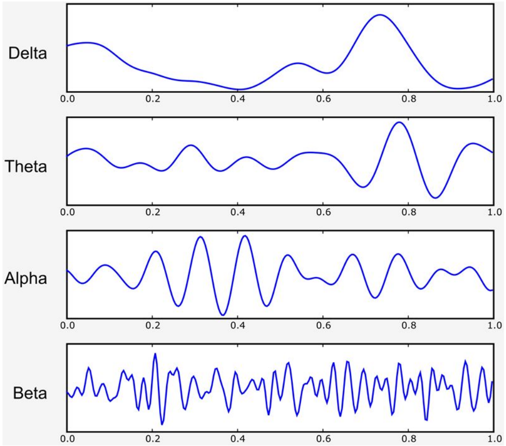
- Filtering
  - Separate frequency bands
- Artefact removal
  - Blind Source Separation (BSS)
- Fourier Transform/Fast Fourier Transform

Frequency bands

- Delta: 1 - 4 Hz
  - Highest amplitude
  - Presents during deep sleep
- Theta: 4 - 8 Hz
  - Focused attention, memory recall (frontal)
- Alpha: 8 - 13 Hz
  - Mental and physical relaxation (occipital, parietal)

Frequency bands (Cont.)

- Beta: 13 - 30 Hz
  - Active thinking, active concentration
- Gamma: < 30 Hz
  - Binding of different populations of neurons



Artifacts

- EOG, EMG, ECG interferes the EEG signals
- Impact of power line noise
- Head swaying and swinging

To remove

- Record other biosignals as well
- Filtering
- Record gyroscope data

Feature extraction

- Power Spectral Density (PSD)
- Common Spatial Patterns (CSP)
- Autoregressive Coefficients (AR)
- Frontal Asymmetry
- Inter-Hemispheric Power Difference
- Inter-Hemispheric Linear Complexity
- Spectral Coherence
- Peak frequency

Pattern recognition/Machine Learning

- Linear Discriminant Analysis
- Quadratic Discriminant Analysis
- Support Vector Machine (Linear, Quadratic, Cubic, Gaussian)
- k-Nearest Neighbour
- Deep Learning techniques

References

- IMOTION EEG Pocket Guide (url: https://imotions.com/eeg-guide-ebook/)
- Wikipedia
- https://www.youtube.com/watch?v=Zd9WhJPa2Ok
- https://youtu.be/hLjxMjBlB9k
- https://www.youtube.com/watch?v=yamA5yM2h0Q
- https://www.youtube.com/watch?time_continue=87&v=tc82Z_yfEwc
- https://youtu.be/wKDimrzvwYA
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- https://youtu.be/jeLghZ8GASA
- www.emotiv.com
- www.choosemuse.com

Other interesting stuffs

- rTMS, tDCS
- Neuromorphic chips
- Quantum Computers (ibm.biz/qx-introduction)
- Augmented Reality (AR) and Virtual Reality (VR) with HMDs and smart glasses

Thanks!

Any questions?

Blog: www.i-linkz.net